

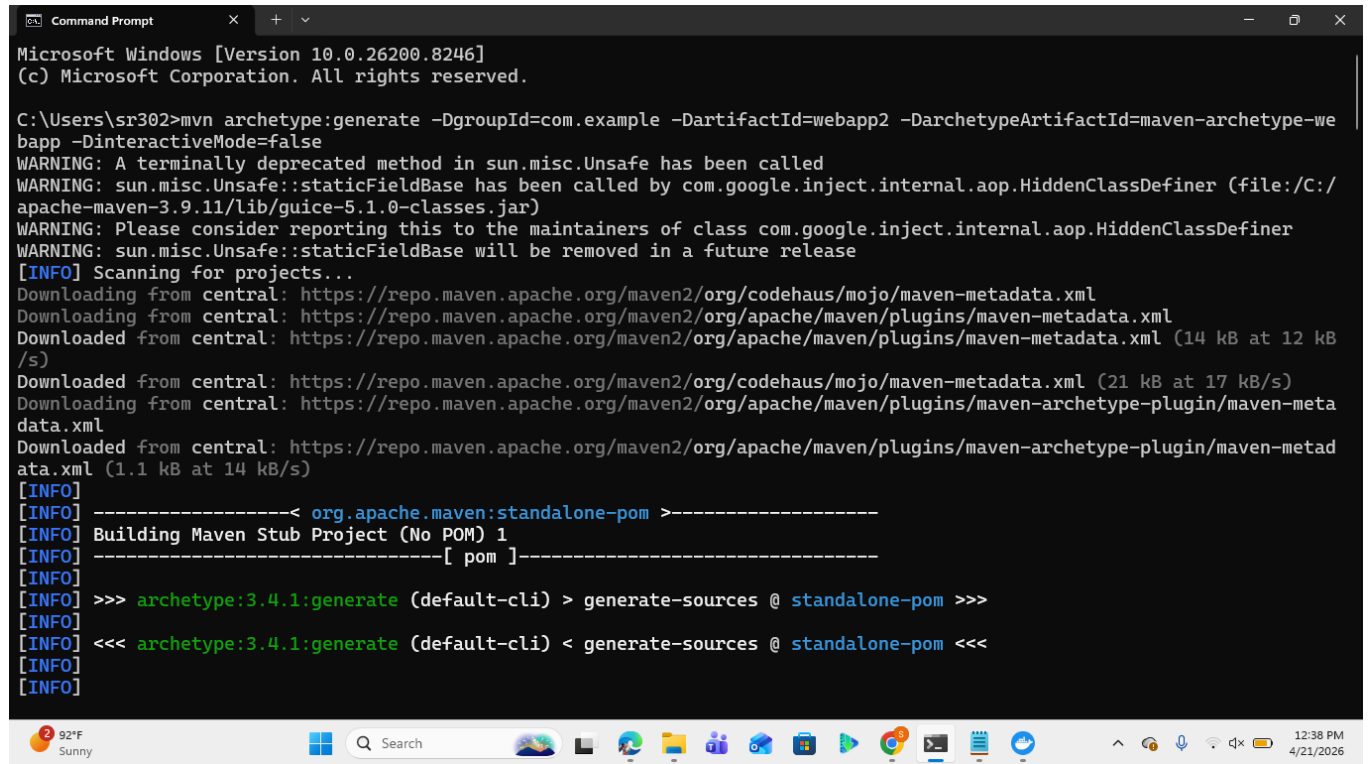
INT-332 Practical-5

Maven Practical

Exercise 1: Maven + Docker Container.

Step 1: Create Maven Project

```
mvn archetype:generate -DgroupId=com.example -DartifactId=my-webapp2 -DarchetypeArtifactId=maven-archetype-webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
```



```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sr302>mvn archetype:generate -DgroupId=com.example -DartifactId=webapp2 -DarchetypeArtifactId=maven-archetype-we
bapp -DinteractiveMode=false
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/
apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 12 kB
/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 17 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metad
ata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-metad
ata.xml (1.1 kB at 14 kB/s)
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
```

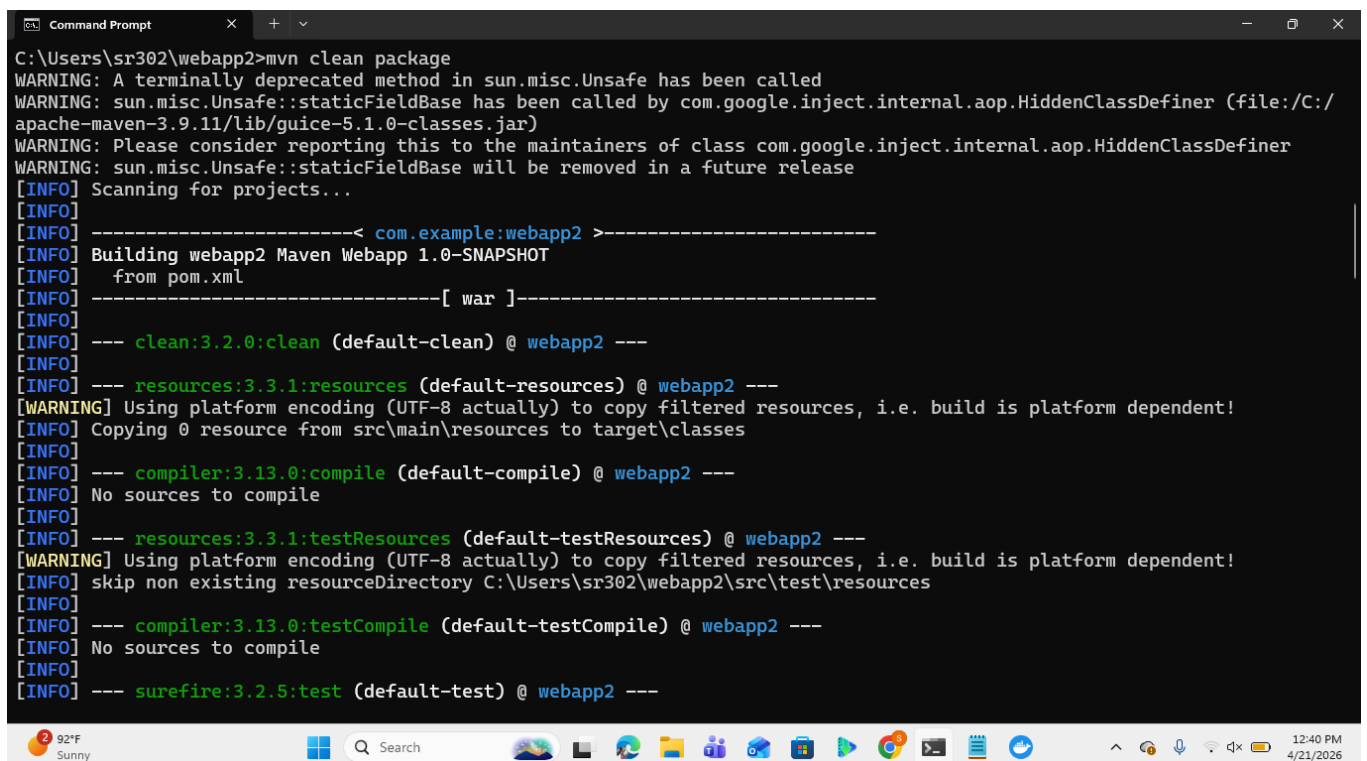
```

[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 19 MB/s)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO]
[INFO] Parameter: basedir, Value: C:\Users\sr302
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: webapp2
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\sr302\webapp2
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 7.615 s
[INFO] Finished at: 2026-04-21T12:24:45+05:30
[INFO]

```

Step 2: Package Application:-

Mvn clean package



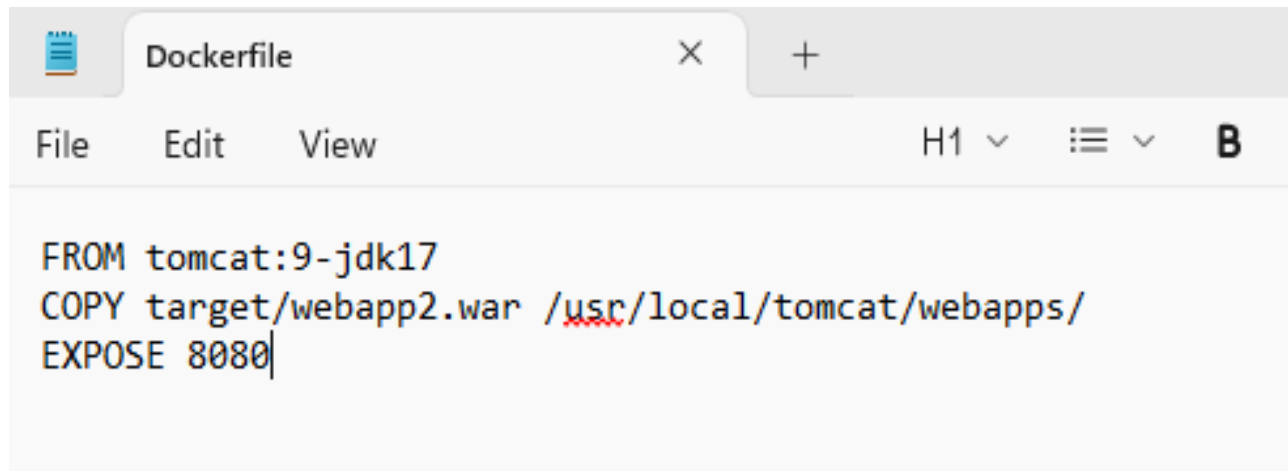
```

C:\Users\sr302\webapp2>mvn clean package
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/
apache-maven-3.9.11/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:webapp2 >-----
[INFO] Building webapp2 Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ webapp2 ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ webapp2 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ webapp2 ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ webapp2 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sr302\webapp2\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ webapp2 ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ webapp2 ---

```

```
[INFO] --- surefire:3.2.5:test (default-test) @ webapp2 ---  
[INFO] No tests to run.  
[INFO] --- war:3.4.0:war (default-war) @ webapp2 ---  
[INFO] Packaging webapp  
[INFO] Assembling webapp [webapp2] in [C:\Users\sr302\webapp2\target\webapp2]  
[INFO] Processing war project  
[INFO] Copying webapp resources [C:\Users\sr302\webapp2\src\main\webapp]  
[INFO] Building war: C:\Users\sr302\webapp2\target\webapp2.war  
[INFO] -----  
[INFO] BUILD SUCCESS  
[INFO] -----  
[INFO] Total time: 4.373 s  
[INFO] Finished at: 2026-04-21T12:25:11+05:30  
[INFO] -----
```

Step 3: Dockerfile :-



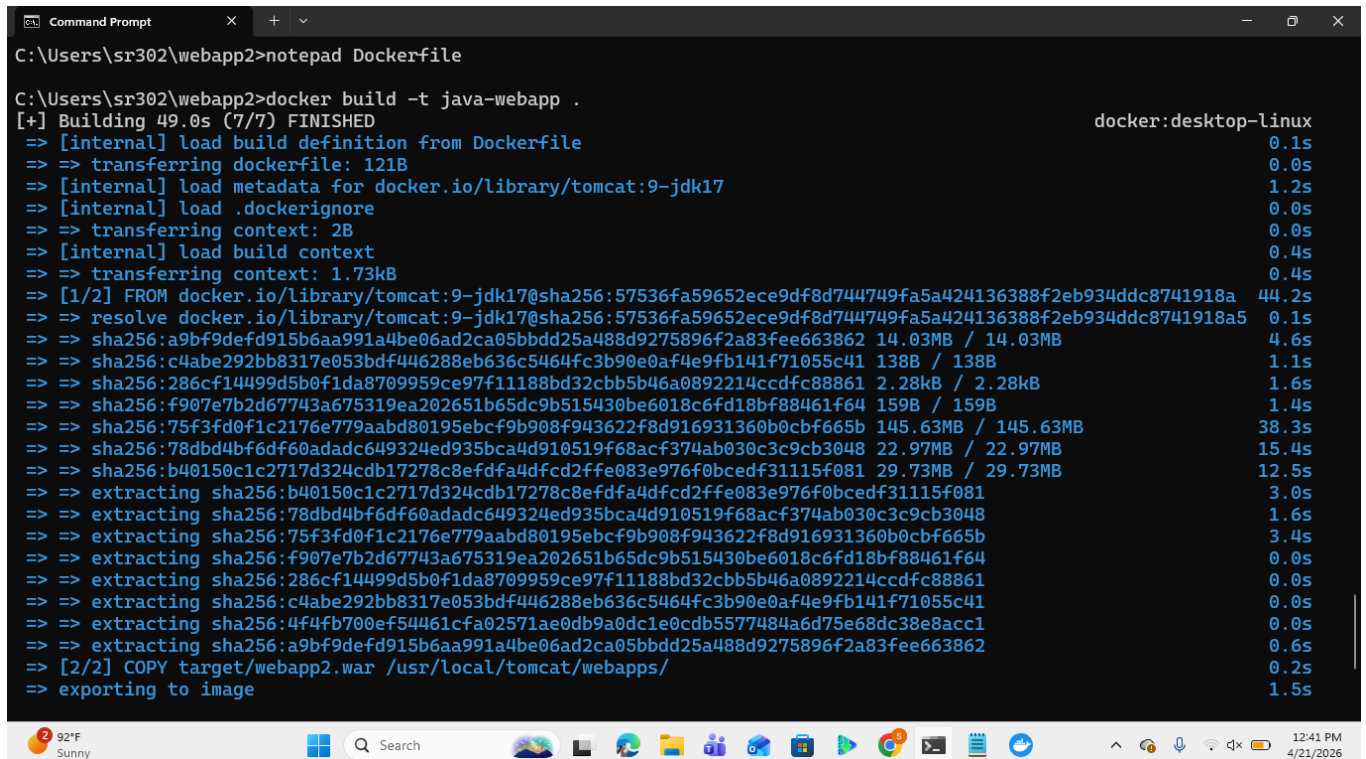
```
FROM tomcat:9-jdk17  
COPY target/webapp2.war /usr/local/tomcat/webapps/  
EXPOSE 8080
```

Step 4: Add dockerfile-maven-plugin in pom.xml:-



```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>webapp2</artifactId>
  <packaging>war</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>webapp2 Maven Webapp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <finalName>webapp2</finalName>
  </build>
</project>
```

Step 5: Run Docker Container:-



```
C:\Users\s302\webapp2>notepad Dockerfile

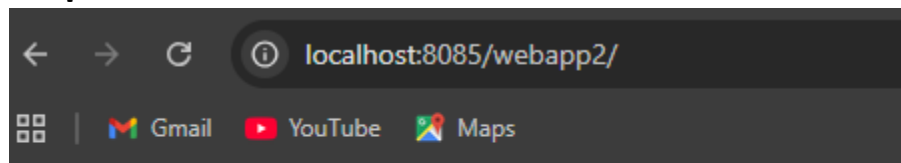
C:\Users\s302\webapp2>docker build -t java-webapp .
[+] Building 49.0s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 121B
=> [internal] load metadata for docker.io/library/tomcat:9-jdk17
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.73kB
=> [1/2] FROM docker.io/library/tomcat:9-jdk17@sha256:57536fa59652ece9df8d744749fa5a424136388f2eb934ddc8741918a5
=> => resolve docker.io/library/tomcat:9-jdk17@sha256:57536fa59652ece9df8d744749fa5a424136388f2eb934ddc8741918a5
=> => sha256:a9bf9defd915b6aa991a4be06ad2ca05bbdd25a488d9275896f2a83fee663862 14.03MB / 14.03MB
=> => sha256:c4abe292bb8317e053bdf446288eb636c5464fc3b90e0af4e9fb141f71055c41 138B / 138B
=> => sha256:286cf14499d5b0f1da8709959ce97f11188bd32cbb5b46a0892214ccdfc88861 2.28kB / 2.28kB
=> => sha256:f907e7b2d67743a675319ea202651b65dc9b515430be6018c6fd18bf88461f64 159B / 159B
=> => sha256:75f3fd0f1c2176e779aabd80195ebcf9b908f943622f8d916931360b0cbf665b 145.63MB / 145.63MB
=> => sha256:78dbd4bf6df60adadc649324ed935bca4d910519f68acf374ab030c3c9cb3048 22.97MB / 22.97MB
=> => sha256:b40150c1c2717d324cdb17278c8efdfa4dfcd2ffe083e976f0bcedf31115f081 29.73MB / 29.73MB
=> => extracting sha256:b40150c1c2717d324cdb17278c8efdfa4dfcd2ffe083e976f0bcedf31115f081
=> => extracting sha256:78dbd4bf6df60adadc649324ed935bca4d910519f68acf374ab030c3c9cb3048
=> => extracting sha256:75f3fd0f1c2176e779aabd80195ebcf9b908f943622f8d916931360b0cbf665b
=> => extracting sha256:f907e7b2d67743a675319ea202651b65dc9b515430be6018c6fd18bf88461f64
=> => extracting sha256:286cf14499d5b0f1da8709959ce97f11188bd32cbb5b46a0892214ccdfc88861
=> => extracting sha256:c4abe292bb8317e053bdf446288eb636c5464fc3b90e0af4e9fb141f71055c41
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1
=> => extracting sha256:a9bf9defd915b6aa991a4be06ad2ca05bbdd25a488d9275896f2a83fee663862
=> [2/2] COPY target/webapp2.war /usr/local/tomcat/webapps/
=> exporting to image
```

```
=> [2/2] COPY target/webapp2.war /usr/local/tomcat/webapps/ 0.2s
=> exporting to image 1.5s
=> => exporting layers 0.0s
=> => exporting manifest sha256:96ffb981dcb0e808891f9ce1a5e191a796b330bc315dc14feb035ccd0b9b7f3b 0.0s
=> => exporting config sha256:2258f4cdeafb34003c51697e5588a577361b440a4a6d8d16a01ad8240bbba5ba 0.0s
=> => exporting attestation manifest sha256:cd6a377459c27fca6c2a8cf64b5bc3263a29cb79cd6467ae78905aa3091ddb6c 0.1s
=> => exporting manifest list sha256:57b910b011bf920e2253a2db8ac984cd5594ce76c56f2889bebad7f51dbb1eda 0.0s
=> => naming to docker.io/library/java-webapp:latest 0.0s
=> => unpacking to docker.io/library/java-webapp:latest 0.1s
```

```
C:\Users\sr302\webapp2>docker run -d -p 8085:8080 --name java-container java-webapp
232c4363256c49f501c72da75aa7fc8daef22192a5375d5956a9b3c3f3ef056a
```

```
C:\Users\sr302\webapp2>
```

Step 5: OutPut:



Hello World!